

Mini Assessment – Answers

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1. If an API returns 429, what should your program do?

A 429 response means the server received too many requests in a given time and is temporarily rejecting new ones. The program should not immediately retry, since that can worsen the situation. Instead, it should pause and retry after a delay, ideally reading the server's Retry-After header if provided. If no such header exists, an exponential backoff strategy should be used, increasing the wait time with each retry.

2. If `response.json()` fails, what should your program do?

If `response.json()` fails, it usually means the response body is not valid JSON, perhaps due to an error page, empty response, or unexpected content type. The program should catch this exception rather than letting it crash the application. It can first check the response status code and content-type header to understand what was actually returned. Logging the raw response text is useful for debugging. Finally, the program should return a clear, appropriate error message to the caller instead of failing silently.

3. If a JSON key is missing, why might `.get()` be safer than `[]`?

Using square brackets to access a dictionary key raises a `KeyError` if that key does not exist, which can crash the program if not handled carefully. The `.get()` method instead returns `None`, or a specified default value, when the key is missing, allowing the program to continue running smoothly. This makes `.get()` especially useful when working with external API responses, where fields may be optional or inconsistently present. It leads to more robust and predictable code with fewer unexpected exceptions.